

Z- Test

Aim:

To perform a *Z-test* in Python to determine whether the average weight of a species of birds differs significantly from the hypothesized population mean of 150 grams.

Procedure:

1. Import the required libraries — NumPy for numerical computation and SciPy for statistical functions.
2. Define the population mean (150 grams) and the significance level ($\alpha = 0.05$).
3. Collect or define a random sample of bird weights (e.g., 30 samples).
4. Calculate the sample mean and sample standard deviation using NumPy functions.
5. Determine the sample size (n).
6. Compute the *Z-statistic* using the formula:

$$Z = \frac{\bar{X} - \mu}{(s/\sqrt{n})}$$

where $\bar{X}$ is the sample mean, μ is the population mean, and s is the sample standard deviation.

7. Find the *p-value* using the standard normal cumulative distribution function.
8. Print the sample mean, *Z-statistic*, and *p-value*.

9. Compare the p -value with α (0.05) to decide whether to reject or fail to reject the null hypothesis.
 10. Conclude whether the mean bird weight differs significantly from 150 grams.
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Program:

```
import numpy as np
from scipy import stats
population_mean = 150
alpha = 0.05
sample = np.array([152, 148, 149, 151, 153, 147, 150, 154, 149, 151,
                  155, 152, 148, 150, 149, 153, 151, 150, 152, 147,
                  149, 150, 151, 152, 148, 150, 149, 151, 153, 150])
sample_mean = np.mean(sample)
sample_std = np.std(sample, ddof=1)
n = len(sample)
z = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
p_value = 2 * (1 - stats.norm.cdf(abs(z)))
print("Sample Mean:", sample_mean)
print("Z-Statistic:", z)
print("P-Value:", p_value)
if p_value < alpha:
    print("Reject the null hypothesis")
```

else:

```
print("Fail to reject the null hypothesis")
```

Output:

Sample Mean: 150.46666666666667

Z-Statistic: 1.2700156853715165

P-Value: 0.20407904295551949

Fail to reject the null hypothesis

Result:

Hence, a Python program was written to perform a *Z-test*, and the result shows that there is **no significant difference** between the average weight of the birds and the population mean of 150 grams at a 5% significance level.